LOST ENERGY

Game Design Document

Third-Person Adventure / Resource Management

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Engine : Unity 6 (URP)

Target Platform : Windows (PC)

Document Version: 1.0
Project Version: 1.0

Lost Energy
A broken world. A fading reality. A loop you can't escape.

Figure 1. *Lost Energy – Main Cover Art*

Figure 2. *Alternative cover design supporting the game's atmospheric and narrative tone.*

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1. Design History

Version 1.0

This version represents the submission-ready current design document for the Lost Energy project.

- Three playable scenes completed: SampleScene, SampleScene2, SampleScene3.
- Third-person character controller, orbit camera, and interaction system integrated.
- Oxygen-based resource management, crystal collection loop, and hazard zone system completed.
- Main menu, pause menu, settings panel, and loading screen implemented.
- NPC dialogue system, inter-scene transition flow, and basic audio management completed.
- Asset, source, and license information integrated into the document.

2. Game Overview

Game Definition

Lost Energy is a single-player third-person adventure game. The player manages their oxygen to survive on a fragmented island floating in the void, collects crystals scattered throughout the environment, avoids hazardous zones, and opens gates that enable progression between scenes.

Figure 3. *General gameplay view showing the player character, HUD, and scene layout together.*

Design Goals

- Establish gameplay that is quickly understood through simple rules yet creates pressure.
- Use the oxygen mechanic simultaneously as a life indicator, time pressure, and strategic resource.
- Combine environmental threats, exploration, and storytelling into a short but cohesive experience.
- Produce a finished-feeling vertical slice within a manageable scope for academic submission.

Core Design Principles

1. Constant Pressure

The player loses oxygen even in safe zones. As a result, inactivity is penalized, decision-making is accelerated, and every crystal becomes meaningful.

2. Clear Objectives

In the first two scenes, the crystal target is explicit. The player can always see what they need to collect, when the door will open, and what the progression condition is.

3. Environmental Storytelling

Red hazardous surfaces, scene transitions, island boundaries, and guardian dialogues work together to reinforce the broken structure of the world.

Core FAQ

Question	Answer
What is the game?	A short-form third-person adventure built around resource management, exploration, and scene-based progression.
Where does it take place?	An island system that is part of an incomplete creation, floating in the void.
What does the player control?	A single character that moves, jumps, interacts, collects crystals, and progresses between scenes.
What is the main focus?	Choosing the right route under oxygen pressure, collecting crystals, and reaching the exit.
What makes it unique?	The oxygen system simultaneously determines time pressure, route optimization, and reward structure.

3. Feature Set

Core Gameplay Features

- Third-person character controller
- Freely rotating orbit camera system
- Real-time oxygen consumption and replenishment system
- Crystal collection and objective completion structure
- IInteractable-based interaction architecture
- Additional oxygen loss from hazard zones
- Death/respawn flow tied to falling and oxygen depletion
- NPC dialogue system and scene-bound narrative delivery

Supporting Systems

- Main menu and pause menu
- Settings panel and volume control
- Scene transition via loading screen
- Scene-based music switching
- VFX and SFX support for crystal collection
- Branching ending choice in the final scene

User Experience Features

- Continuously visible oxygen and crystal indicators
- Prompt text on screen for nearby interaction targets
- Clear control scheme
- Compact level design suited to short play sessions

4. Game World

General Structure

The Lost Energy world is designed as an incomplete realm of existence. The island is physically bounded; going beyond its borders results in death. Each new scene represents a zone where deterioration has become more visible.

World Features

Fragmented Island Structure

The play area is built on a layout where safe surfaces coexist with risky gaps. This makes navigation, staying on platforms, and selecting safe routes important.

Deterioration Zones

Red hazardous surfaces do not merely provide visual variety; they are mechanical threat zones that force the player to alter their route.

Physical World Summary

Parameter	Value
Perspective	Third-person (3D)
Walk Speed	10 units/sec
Sprint Speed	25 units/sec
Jump Height	1.5 units
Camera Distance	5 units

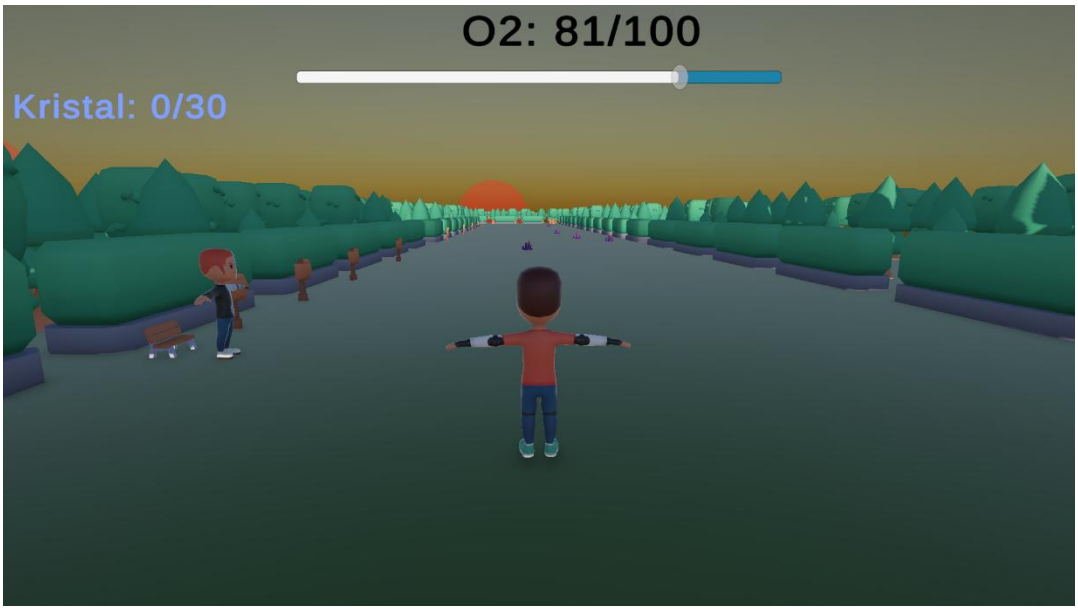

Figure 4. *General view of Scene 2 showing increased environmental deterioration.*

Figure 5. *General environment view belonging to the final scene (Scene 3).*

5. Camera and Controls

Camera System

The camera system is managed within PlayerController3P. The camera orbits around the player; yaw and pitch inputs are handled independently. Since movement direction is calculated relative to the camera's view, the control feel is designed to resemble third-person action games.

Parameter	Value
Mouse Sensitivity	0.2
Pitch Minimum	-30°
Pitch Maximum	+60°
Camera Distance	5 units

Control Scheme

Action	Input
Move	WASD / Arrow Keys
Jump	Space
Sprint	Left Shift
Interact	E
Camera	Mouse
Pause / Release Cursor	ESC

6. Gameplay Systems

Oxygen System

Oxygen is the game's primary resource structure. The player consumes oxygen as long as they are present in the scene. When the value reaches zero, the character dies and returns to the last safe starting point.

Zone	Starting Oxygen	Normal Consumption	Hazard Extra	Crystal Bonus
Scene 1	100	3 units/sec	+5 units/sec	+5
Scene 2	100	3 units/sec	+6 units/sec	+6

Crystal System

- Crystals are spawned in defined areas within the scene.
- The player collects a crystal by interacting with the E key.
- Visual and auditory feedback is generated during collection.
- In the first two scenes, completing the target crystal count opens the exit door.
- In the final scene there is no crystal target; the final choice is presented directly.

Hazard Zone System

- HazardZone areas increase oxygen consumption.
- These zones are visually marked with red surfaces.
- When entering a hazard area, music and tension are intensified.

Figure 6. *Example screenshot showing the effect of a hazard zone on the play area.*

Figure 7. *Second view showing the pressure a hazard zone exerts on the player.*

Interaction System

The interaction structure is built on the IInteractable interface. Crystals, doors, NPCs, and potions all use the same approach. PlayerInteraction identifies the most suitable target for the player and displays the appropriate prompt text on screen.

Figure 8. *The interaction prompt and HUD components visible during crystal collection.*

Reward and Progression Loop

```
Collect Crystal --> Replenish Oxygen --> Survive Longer
                --> Collect More Crystals --> Complete Objective
                --> Open Exit Door --> Transition to Next Scene
```

7. World Layout and Progression Flow

General Flow

The game consists of three main scenes. In the first two scenes the player collects a set number of crystals; in the third scene the experience ends or loops through a final choice.

Scene 1 – Island Entrance

Field	Description
Atmosphere	Relatively calm starting zone
Crystal Target	25
Hazards	Falling off island edges, low-density hazard zones
NPC	Guardian
Exit	ExitDoorSceneLoader → Scene 2

Scene 2 – Deteriorated Zone

Field	Description
Atmosphere	Increased deterioration with more aggressive risk areas
Crystal Target	30
Hazards	Falling off island edges, dense hazard zones, stronger route pressure
NPC	Guardian
Exit	ExitDoorSceneLoader → Scene 3

Scene 3 – End of Creation

Field	Description
Atmosphere	The final threshold of an unfinished existence
Crystal Target	0
Hazards	Final decision: loop or exit choice
NPC	Guardian
Exit	FinalDoorController → Loop end or Win screen

Game Flow Diagram

```
Main Menu
 --> LoadingScreen
 --> Scene 1  (Crystal Target: 25)
 --> Scene 2  (Crystal Target: 30)
 --> Scene 3  (No crystal target)
     --> Loop Door  : Return to Scene 1
     --> Exit Door  : Congratulations Screen
```

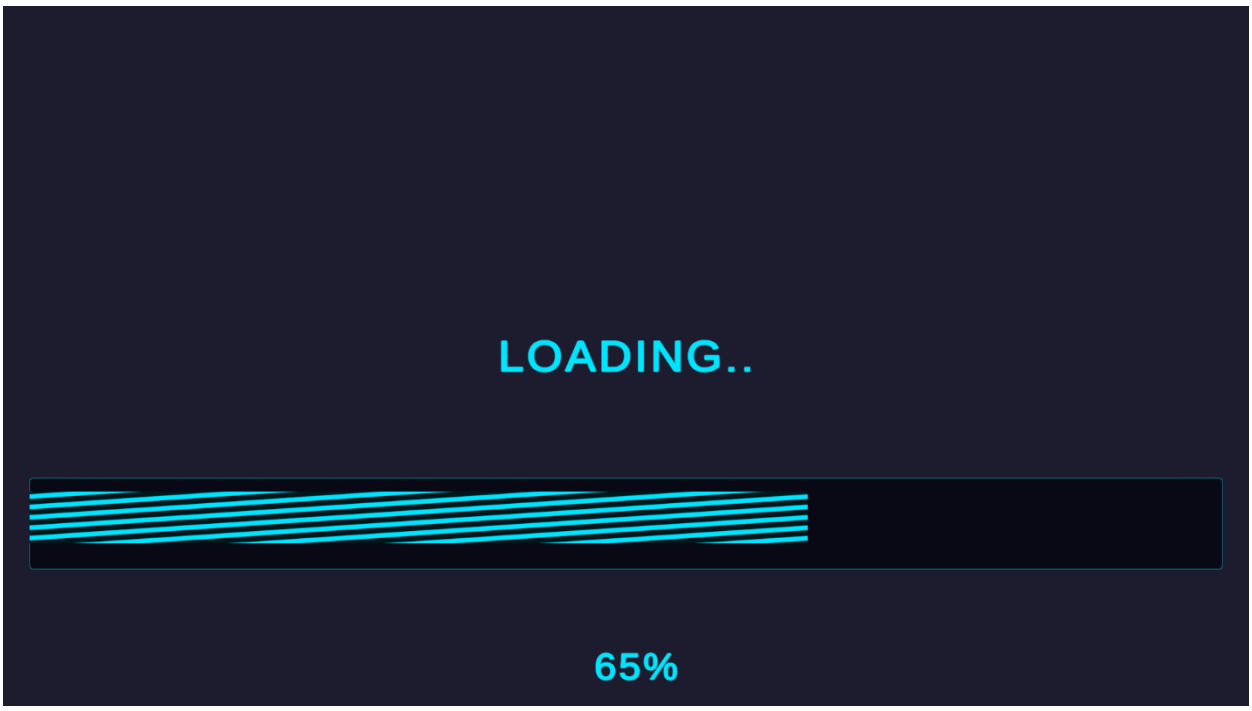

Figure 9. *The loading screen is used for transitions between scenes.*

Figure 10. *The loop-return door interaction in the final scene.*

8. Characters

Player Character

Attribute	Description
Model	Kenney Animated Characters Protagonists
Control Setup	PlayerController3P + CharacterController
Primary Goal	Collect crystals and reach the exit under oxygen management
Death Conditions	Oxygen reaching zero or falling so that Y < -1

NPC – Guardian

The Guardian is the game's sole narrative-focused NPC. It greets the player in each scene, provides information about the current zone, and gradually reveals the world's background. Because the character has merged with the island, it cannot leave — a fact that reinforces its narrative function.

Dialogue System Behavior

- Managed through NPCDialogue and DialogueData structures.
- Oxygen consumption is paused while dialogue is active.
- Dialogue simultaneously provides system tutorials and story delivery.

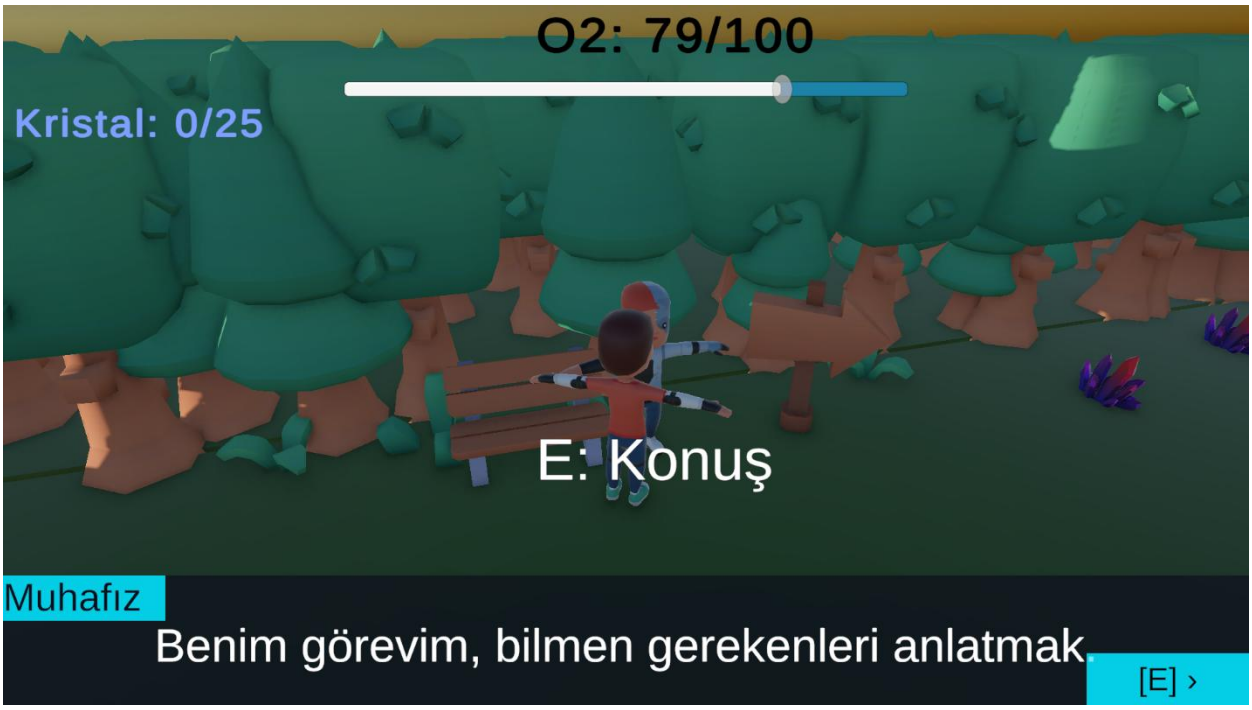

Figure 11. *The dialogue panel that opens during interaction with the Guardian character.*

9. User Interface

HUD Components

Component	Function
Oxygen Slider	Visually displays the current oxygen level
Oxygen Value	Shows the numerical equivalent of oxygen
Crystal Counter	Shows collected crystal count in current / target format
Interaction Prompt	Provides contextual information for nearby interactable objects

Panel Structure

Panel	Trigger Condition
Game Over	Oxygen depletion or death by falling
Win	Selecting the winning exit in the final scene
Pause	ESC key
Settings	Accessed from within the Pause menu
Controls	Accessed from the main menu or Pause menu
Loading Screen	Opens automatically during scene transitions

Menu Structure

The main menu contains options to start the game, access settings, view controls, and quit. At game start Time.timeScale = 1f and the cursor is free.

Figure 12. *The game's main menu interface.*

Figure 13. *The UI screen showing the game's control scheme.*

Figure 14. *The settings panel containing volume level and mute options.*

10. Audio and Music

Audio Architecture

The project uses three primary channels through GameAudioMixer.

Channel	Purpose
MasterVol	Overall level of all sounds
MusicVol	Background music tracks
SFXVol	In-game effects, dialogue audio, and hazard warnings

Music Usage

- Different music streams are used for the main menu and gameplay scenes.
- Hazard zones are supported with additional audio layers that heighten tension.
- MusicManager maintains music continuity across scene transitions.

Source Summary

Source	Content
Flowerhead – SomeWhatGood: Lofi	Observatory and chill 2
Not Jam Music Pack	ChillMenu, CriticalTheme, SwitchWithMeTheme
Freesound	Crystal collection and NPC sound effects

11. Single-Player Experience

Game Loop

The single-player experience is built on a scene-based structure that creates short but intense decision pressure. In each scene the player reads the environment, evaluates risky routes, collects crystals, and reaches the exit door.

Estimated Playtime

Play Style	Duration
Speed Completion	5–6 minutes
Exploration and Dialogue Focus	10–12 minutes

Win and Lose Conditions

State	Condition
Win	Selecting the exit ending via FinalDoorController in Scene 3
Lose	Oxygen reaching zero or falling from the scene

Figure 15. *The exit door interaction offering the option to complete the game.*

Figure 16. *The result screen shown after the game is completed.*

12. Story and Narrative

Story Summary

Lost Energy follows a character trapped inside an incomplete creation who searches for a way out. The game world is a realm of existence that cannot fully fulfil its function; as a result, even oxygen is unstable. The red deterioration zones represent the world's dissolving structure. Crystals function as fragments that have remained intact within this broken order.

Thematic Structure

Element	Meaning
Island	An incomplete creation
Oxygen	The world's unstable and incomplete nature
Crystals	Preserved fragments of reality
HazardZone	Physical manifestation of deterioration
Guardian	A witness aware of the system yet unable to escape it

Scene-Based Narrative Development

Scene	Narrative Function
Scene 1	Introduction of the world's rules and the primary objective
Scene 2	Intensification of deterioration and danger
Scene 3	Revelation of the truth and presentation of a choice to the player

Character Motivations

- The player character tries to survive and find the way out before understanding the world they are in.
- The Guardian serves a guiding role; it acts more from a sense of duty than hope.

13. Technical Implementation

Architecture Summary

```
LostEnergy Namespace
├── GameManager
├── MusicManager
├── SettingsManager
├── SceneLoader
└── UIManager
```

Core Scripts

Script	Responsibility
PlayerController3P	Character movement, sprint, jumping, and orbit camera
OxygenSystem	Oxygen consumption and replenishment
GameManager	Crystal counting, objective checking, and death flow
CrystalCollectible	Crystal interaction, SFX and VFX triggering
HazardZone	Applying additional oxygen consumption
DialogueManager	Dialogue flow and oxygen pause
RespawnManager	Position reset after death
PauseManager	Pause flow and panel management
LoadingScreenManager	Asynchronous scene loading and loading screen

Technical Decisions

- Game logic is managed through modular script structures.
- MusicManager uses the DontDestroyOnLoad approach for persistence across scenes.
- SettingsManager stores audio settings with PlayerPrefs.
- References are cached during Start() to reduce cost within Update().

Performance Notes

- The crystal VFX structure uses an Instantiate + Destroy approach, sufficient for low object density.
- Scene scales are limited for short gameplay sessions, keeping overall runtime cost manageable.

14. Assets and License Information

License Summary

Asset Group	License / Usage Terms
Kenney Assets	CC0 1.0 Universal
Unity Asset Store Assets	Standard Unity Asset Store EULA
Not Jam Music Pack	CC0 1.0
Flowerhead – SomeWhatGood: Lofi	Royalty free (author declaration)
Freesound effects	Per-file CC license
Unity built-in packages	Unity Asset Store EULA / Unity Companion License

Kenney Assets

All Kenney assets are published under the CC0 1.0 Universal license (https://creativecommons.org/publicdomain/zero/1.0/).

Asset Pack	Version	Source
Animated Characters Protagonists	1.1	kenney.nl/assets/animated-characters
Fantasy Town Kit	2.0	kenney.nl/assets/fantasy-town-kit
Graveyard Kit	5.0	kenney.nl/assets/graveyard-kit
Nature Kit	2.1	kenney.nl/assets/nature-kit
Platformer Kit	4.1	kenney.nl/assets/platformer-kit

Unity Asset Store Assets

Asset Pack	Publisher	Version
Stylized Crystal	LowlyPoly	1.0
Stylized door	lowpoly89	1.0

License: Unity Asset Store End User License Agreement – unity.com/legal/as-terms

Audio and Music Sources

File / Pack	Source	License
794489__gobbe57__coin-pickup.wav	freesound.org/people/gobbe57/	Freesound per-file
822698__metris__retro-npc-voice.wav	freesound.org/people/Metris/	Freesound per-file

File / Pack	Source	License
ChillMenu.wav	not-jam.itch.io/not-jam-music-pack	CC0 1.0
CriticalTheme.wav	not-jam.itch.io/not-jam-music-pack	CC0 1.0
SwitchWithMeTheme.wav	not-jam.itch.io/not-jam-music-pack	CC0 1.0
Observatory and chill 2.wav	flowerheadmusic.itch.io/somewhat-good-lofi	Royalty free

Unity Built-in Packages

Package	Publisher	License
TextMesh Pro	Unity Technologies	Unity Asset Store EULA
Input System	Unity Technologies	Unity Companion License
Post Processing Stack	Unity Technologies	Unity Companion License

Usage Notes

- Usage terms for Unity Asset Store assets are covered under the standard Unity EULA.
- For files sourced from Freesound, the license type must be verified per file.
- If a CC BY licensed Freesound file is used, written attribution is required.

15. Installation and Running

Development Environment Requirements

- Unity 6 (recommended: 6000.0.x LTS)
- Universal Render Pipeline (URP)
- Input System package

Minimum System Requirements

Component	Specification
OS	Windows 10 64-bit
CPU	Intel Core i3 class or equivalent
Memory	4 GB RAM
GPU	DirectX 11 compatible graphics card
Storage	At least 1 GB free space
Resolution	1280 x 720

Build Steps

1. Open File → Build Settings inside the Unity Editor.
2. Verify that scenes are listed in the following order:

```
0 – MainMenu
1 – LoadingScreen
2 – SampleScene
3 – SampleScene2
4 – SampleScene3
```

3. Select Windows, x86_64 as the platform.
4. Use Build and select Builds/Windows/ as the output folder.

Running the Game

The compiled build is launched via the Builds/Windows/Lost Energy.exe file.

Quick Control Reference

```
WASD        -> Move
Left Shift  -> Sprint
Space       -> Jump
E           -> Interact / Collect Crystal
Mouse       -> Camera
ESC         -> Pause Menu
```